

9-5V DC-DC Converter to Run Fan Load at Full Speed and Half Speed

with IR2110 Gate Driver

Project Report

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May 7, 2026

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1 Aim

To design, construct, and analyse a PWM-controlled DC–DC Buck (step-down) converter that steps down a 9 V DC supply to a regulated 5 V DC output using an N-Channel MOSFET driven by the IR2110 gate driver IC, with the PWM signal generated by an ESP8266 / Arduino microcontroller, and to verify the operation by driving a 5 V DC motor as the load. The fan load is then run at full speed at 5V and half speed at 2.5V by varying the duty cycle.

2 Abstract

This project presents a compact, microcontroller-driven buck converter designed to step down a 9 V DC input to a stable 5 V DC output suitable for driving a small DC motor.

The switching action is provided by an N-Channel power MOSFET operating at a duty cycle of approximately 55%, controlled by a PWM signal from an ESP8266 (or Arduino).

Because the ESP8266's 3.3 V logic level is insufficient to fully enhance a standard power MOSFET, an IR2110 high- and low-side gate driver IC is employed as an intermediate stage.

The IR2110 receives the low-power PWM signal and produces the high-current, 9 V-swing gate pulses required to switch the MOSFET efficiently.

A bootstrap network (diode + $10\mu F \parallel 100nF$ capacitors) sustains the gate drive voltage during the ON state.

Energy storage and filtering are provided by a $260\mu H$ inductor and a $22\mu F$ output capacitor.

A 1N5819 Schottky diode provides a low-loss freewheeling path during the MOSFET OFF interval.

The resulting converter achieves efficient DC–DC conversion with minimal component count and is well suited to battery-powered embedded applications.

3 Principle of Operation

3.1 Buck Converter Fundamentals

A buck (step-down) converter is a switched-mode power supply (SMPS) topology that converts a higher DC input voltage V_{in} to a lower DC output voltage V_{out} with high efficiency. The relationship between input and output is governed by the duty cycle D :

$$V_{out} = D \times V_{in}$$

where $D = \frac{t_{on}}{T}$ is the fraction of the switching period T during which the switch (MOSFET) is ON. For this design, $V_{in} = 9V$ and the target load is a 5V DC motor.

Duty Cycle Calculations for Speed Control

To control the speed of the motor, the average output voltage is varied by adjusting the PWM duty cycle.

- **Full Speed Operation** ($V_{out} = 5V$): To run the motor at its rated full speed, the duty cycle is calculated as:

$$D_{full} = \frac{V_{out}}{V_{in}} = \frac{5}{9} \approx 0.5556 \text{ (55.6\%)}$$

- **Half Speed Operation** ($V_{out} = 2.5V$): For half-speed operation, the output voltage is reduced to 2.5V. The corresponding duty cycle is:

$$D_{half} = \frac{2.5}{9} \approx 0.2778 \text{ (27.8\%)}$$

3.2 Two Operating Intervals

MOSFET ON ($0 \leq t < t_{on}$)

The MOSFET conducts and current flows from V_{in} through the inductor L to the load, simultaneously charging the output capacitor C .

The inductor stores energy as its current ramps up linearly:

$$\frac{di_L}{dt} = \frac{V_{in} - V_{out}}{L}$$

The Schottky diode is reverse-biased and does not conduct.

MOSFET OFF ($t_{on} \leq t < T$)

The MOSFET turns off and the inductor opposes the sudden change in current by forward-biasing the freewheeling (Schottky) diode.

Current continues to flow through:

$$L \rightarrow load \rightarrow D1 \rightarrow L$$

and the inductor releases its stored energy to maintain the output voltage.

The inductor current ramps down:

$$\frac{di_L}{dt} = \frac{-V_{out}}{L}$$

3.3 Role of the IR2110 Gate Driver

The ESP8266 outputs a 3.3 V PWM signal, which is insufficient to drive a power MOSFET gate directly.

Typical MOSFET threshold:

$$V_{GS(th)} \approx 3V - 4V$$

with full enhancement requiring approximately 10 V.

The IR2110 level-shifts and amplifies this signal to produce a 9 V gate drive, ensuring the MOSFET switches into full enhancement (low $R_{DS(on)}$) and turns off completely, minimising conduction and switching losses.

A bootstrap circuit charges the floating bootstrap capacitor:

$$C_{boot} = 10 \mu F \parallel 100 nF$$

to the gate drive supply voltage during the OFF phase of the switch.

4.2 Stage 2 – Buck Power Stage

- N-Channel MOSFET (Low-Side Switch)
- 1N5819 Schottky Freewheeling Diode
- Power Inductor: $260\ \mu H$
- Output Capacitor: $22\ \mu F$
- Load: 5 V DC Motor

5 Component Summary

Reference	Component	Function
U1	IR2110	PWM amplification
Q1	N-Channel MOSFET	Main power switch
D1	1N5819	Freewheeling diode
L1	$260\ \mu H$	Energy storage
C1	$22\ \mu F$	Output filtering
Rg	$10\ \Omega$	Gate control
R1	$100\ \Omega$	PWM input resistor
MCU	ESP8266	PWM generation
M1	5 V DC Motor	Load

6 Inference

6.1 Voltage Conversion

Experimental results confirm:

$$9 \times 0.55 = 4.95V \approx 5V$$

Minor deviation from ideal value occurs due to:

- Diode forward voltage drop
- MOSFET conduction losses
- Inductor DC resistance

6.2 Role of the IR2110

Without the IR2110, the ESP8266's 3.3 V logic would only partially enhance the MOSFET, causing excessive heat and poor efficiency.

The IR2110 ensures:

- Full MOSFET enhancement

- Faster switching
- Lower losses
- Improved efficiency

6.3 Bootstrap Capacitor Importance

The bootstrap capacitor charges to approximately V_{CC} when the MOSFET is OFF.

During the ON phase, the capacitor supplies the floating gate drive voltage.

Without this network, the gate drive voltage would collapse at higher duty cycles.

6.4 Inductor and Capacitor Ripple

The $260\ \mu H$ inductor keeps the converter in Continuous Conduction Mode (CCM).

The $22\ \mu F$ capacitor reduces output voltage ripple and improves motor stability.

6.5 Schottky Diode Selection

The 1N5819 Schottky diode is preferred because of:

- Low forward voltage drop
- Very fast switching
- Reduced switching losses

7 Conclusion

A functional PWM-controlled buck converter was successfully implemented using:

- ESP8266 microcontroller
- IR2110 gate driver
- N-Channel MOSFET
- 1N5819 Schottky diode
- $260\ \mu H$ inductor
- $22\ \mu F$ capacitor

The circuit successfully steps down a 9 V input to approximately 5 V at nearly 55% duty cycle and powers a 5 V DC motor.

The experimental results closely match theoretical predictions, validating the design methodology and component selection.